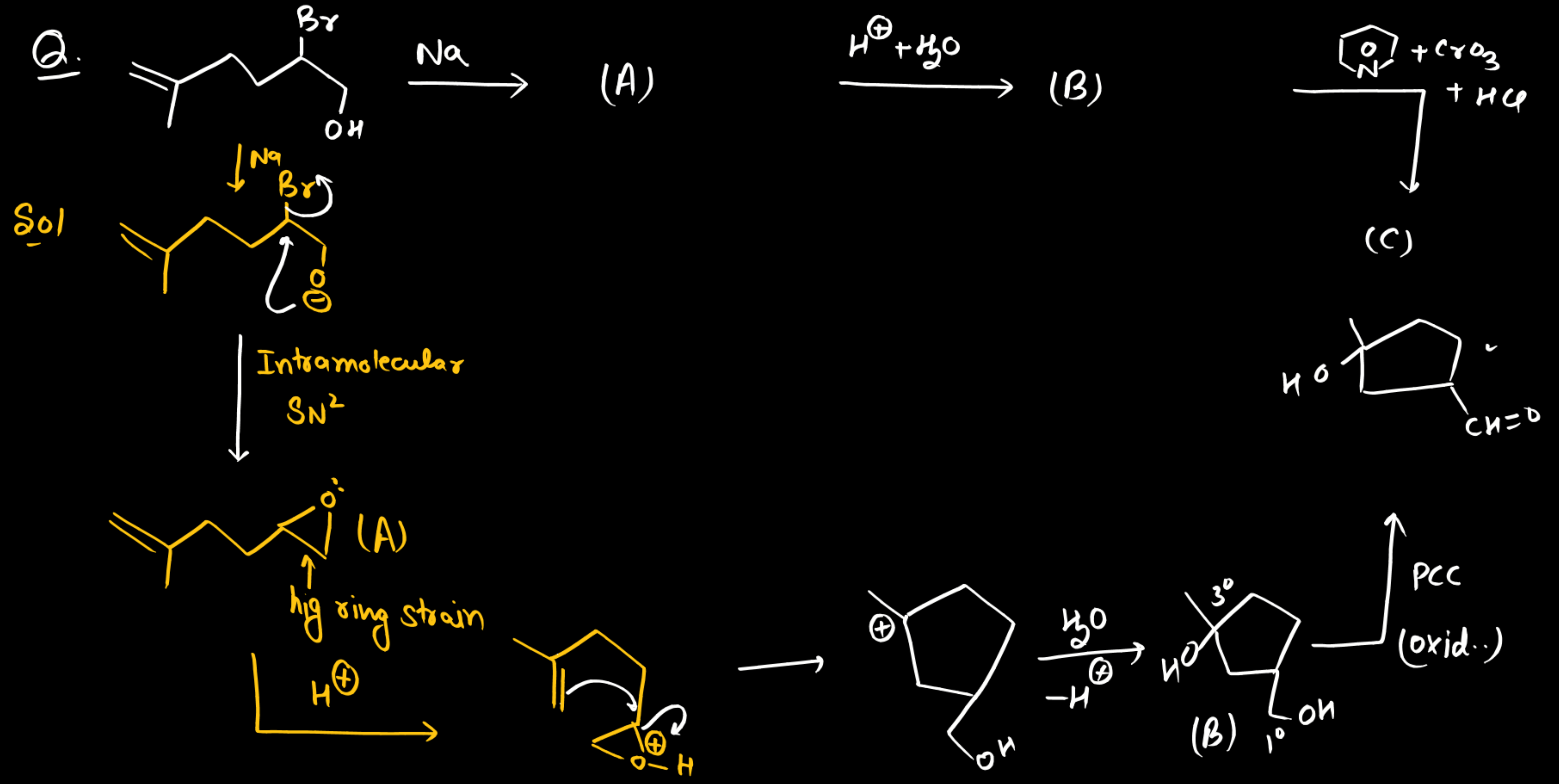
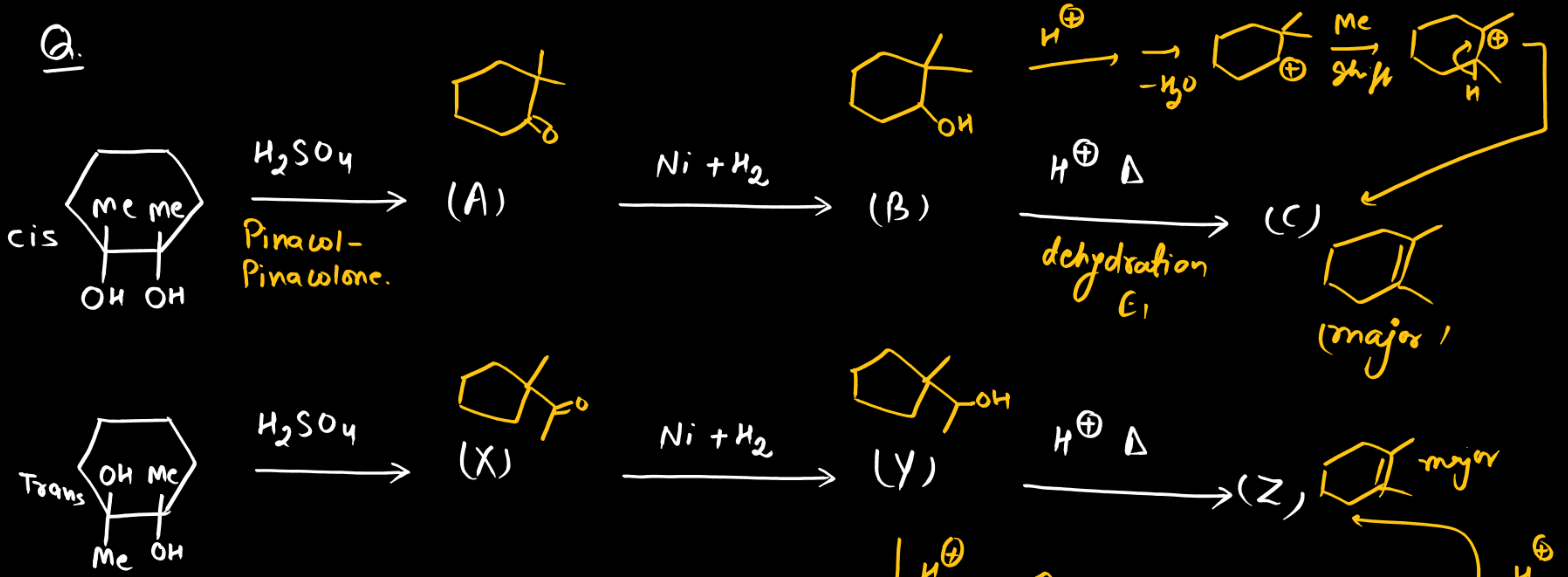
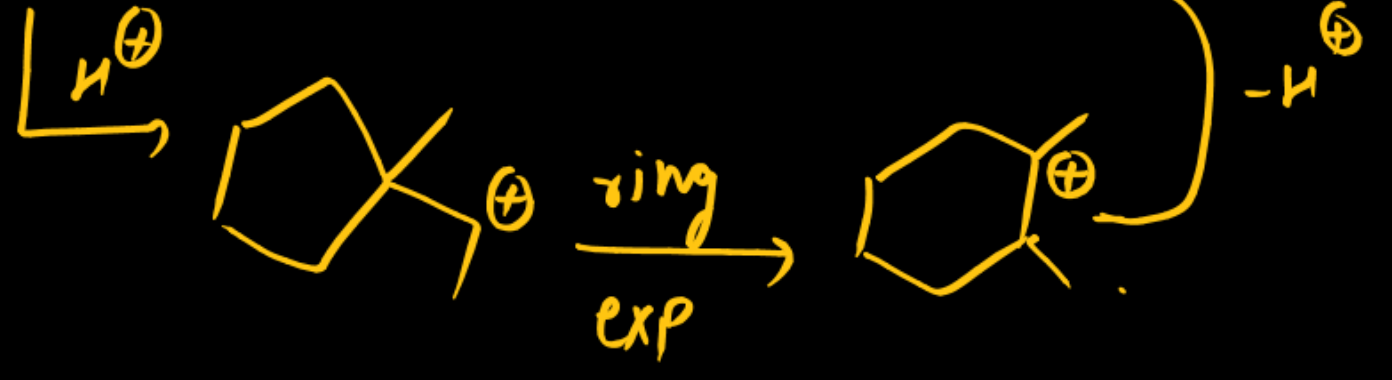
# competishun advanced training session (CATS)



Q.



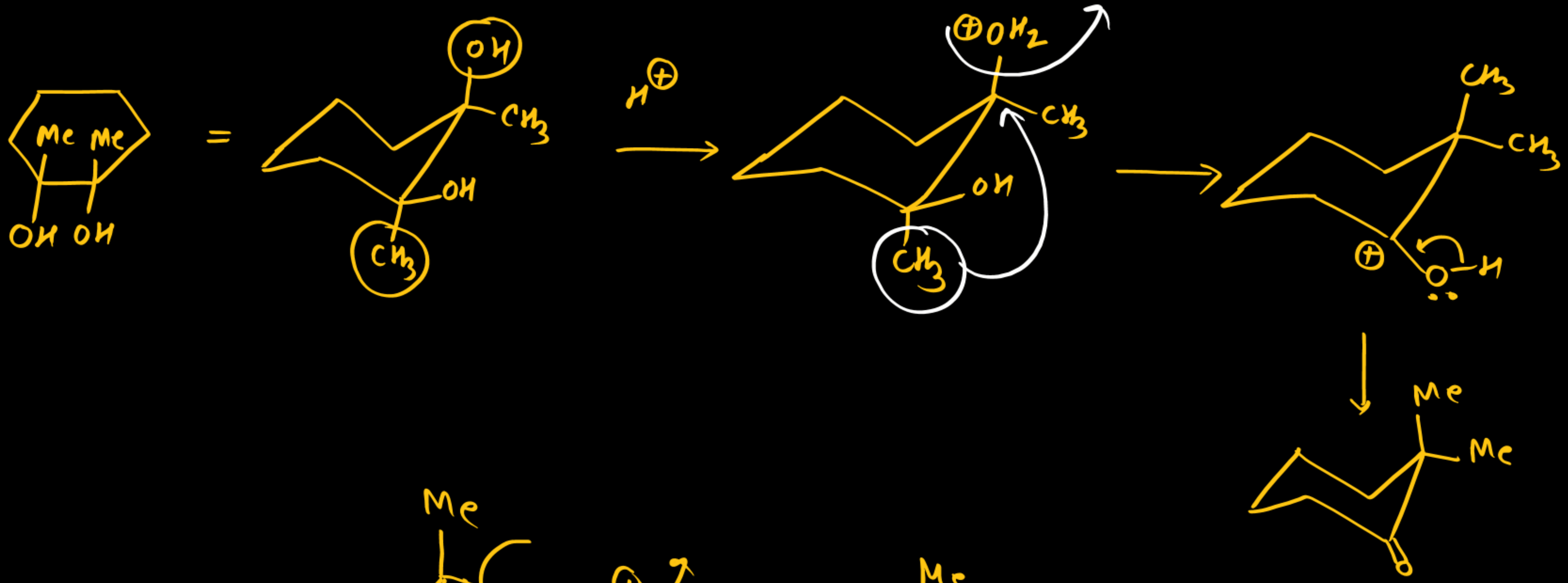




⇒ identify relation between C & Z

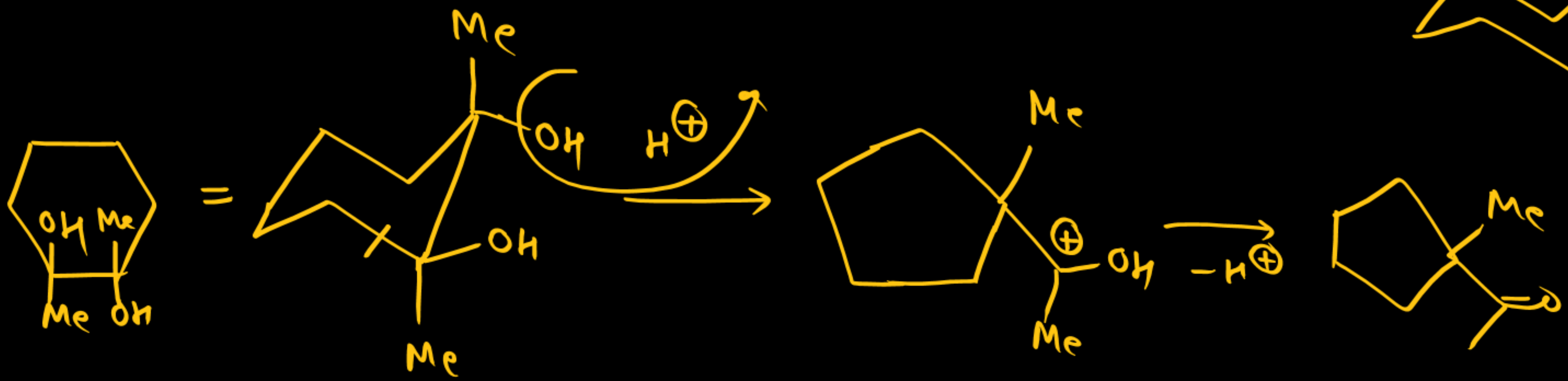
⇒ which can give +ve iodoform test | X, Y

Sol

cis



Trans



# Iodoform test

$$CH_3-\overset{\overset{O}{\parallel}}{C}-H$$

$$CH_3-\overset{\overset{O}{\parallel}}{C}-R \leftarrow \text{alkyl}$$

$$CH_3-\overset{\overset{O}{\parallel}}{C}-Ar$$

$$CH_3-\underset{\underset{H}{|}}{\overset{\overset{OH}{|}}{C}}-(H/R/Ar)$$

$$CH_3-\underset{\underset{OR}{|}}{\overset{\overset{OH}{|}}{C}}-(H/R/Ar)$$

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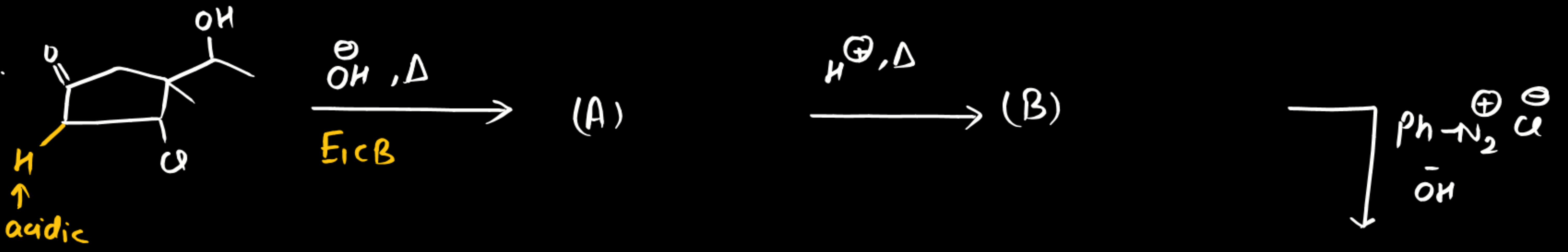
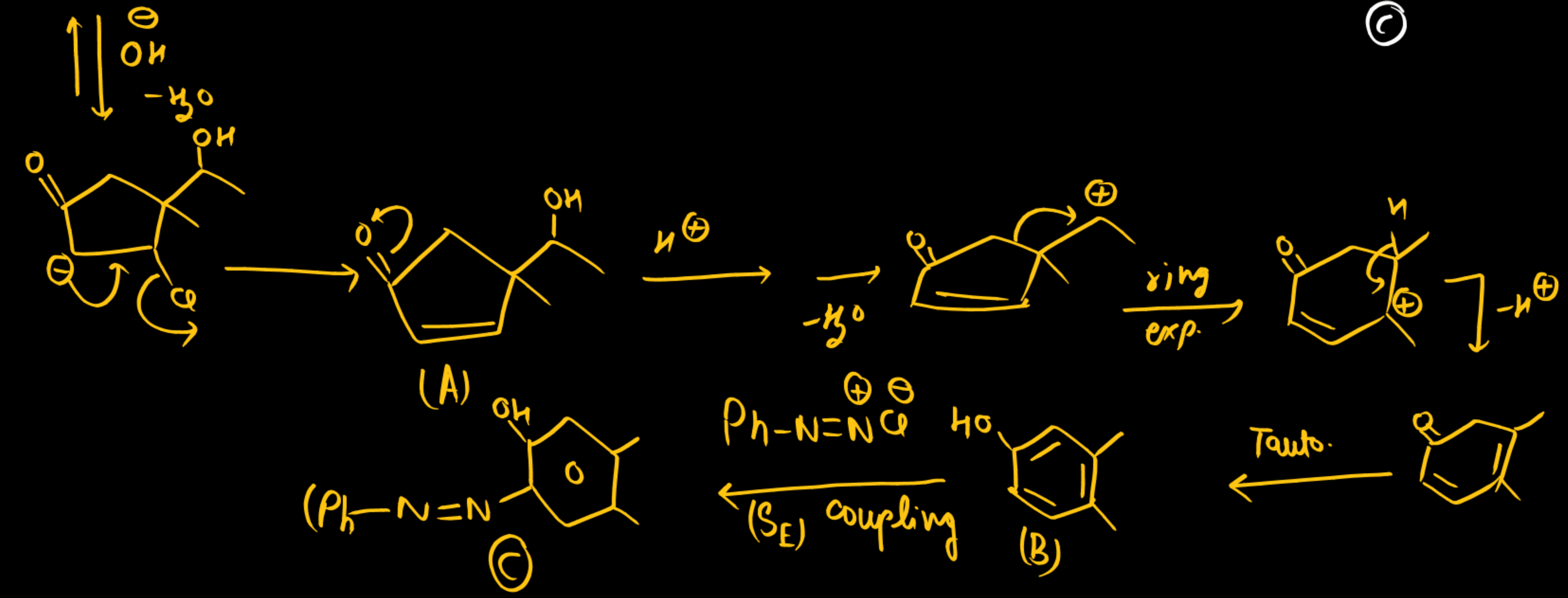
Q.



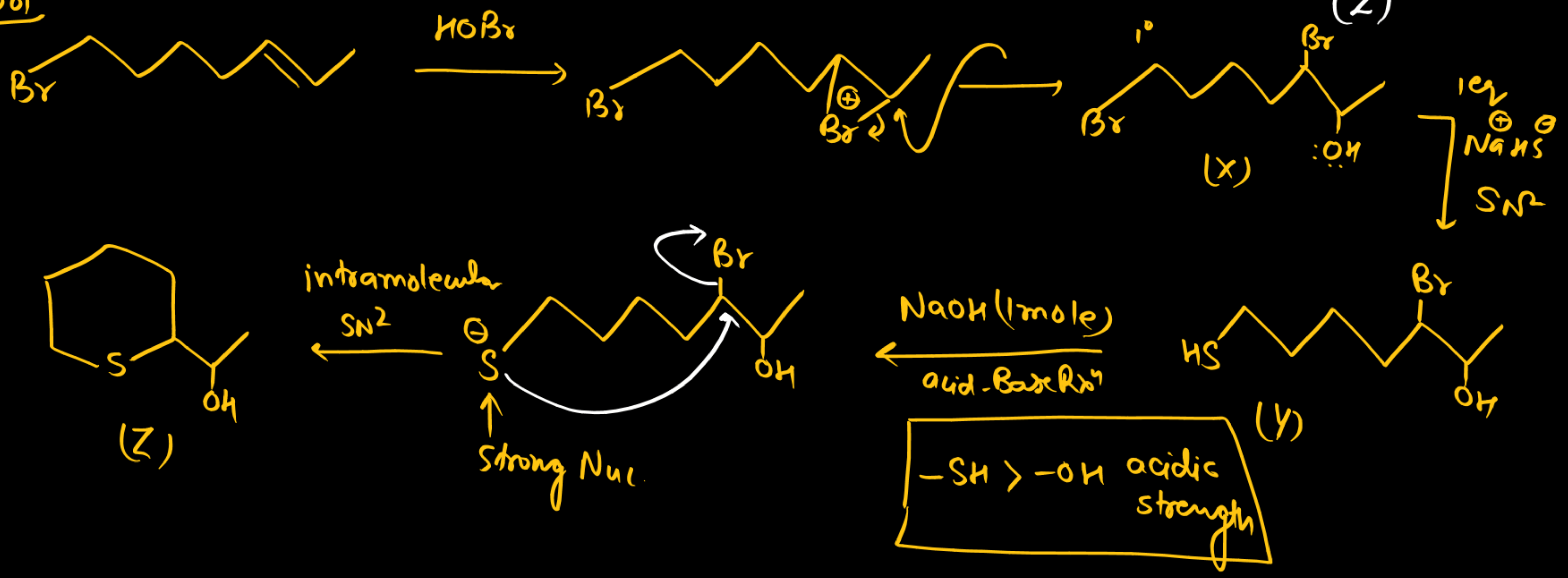

Sol



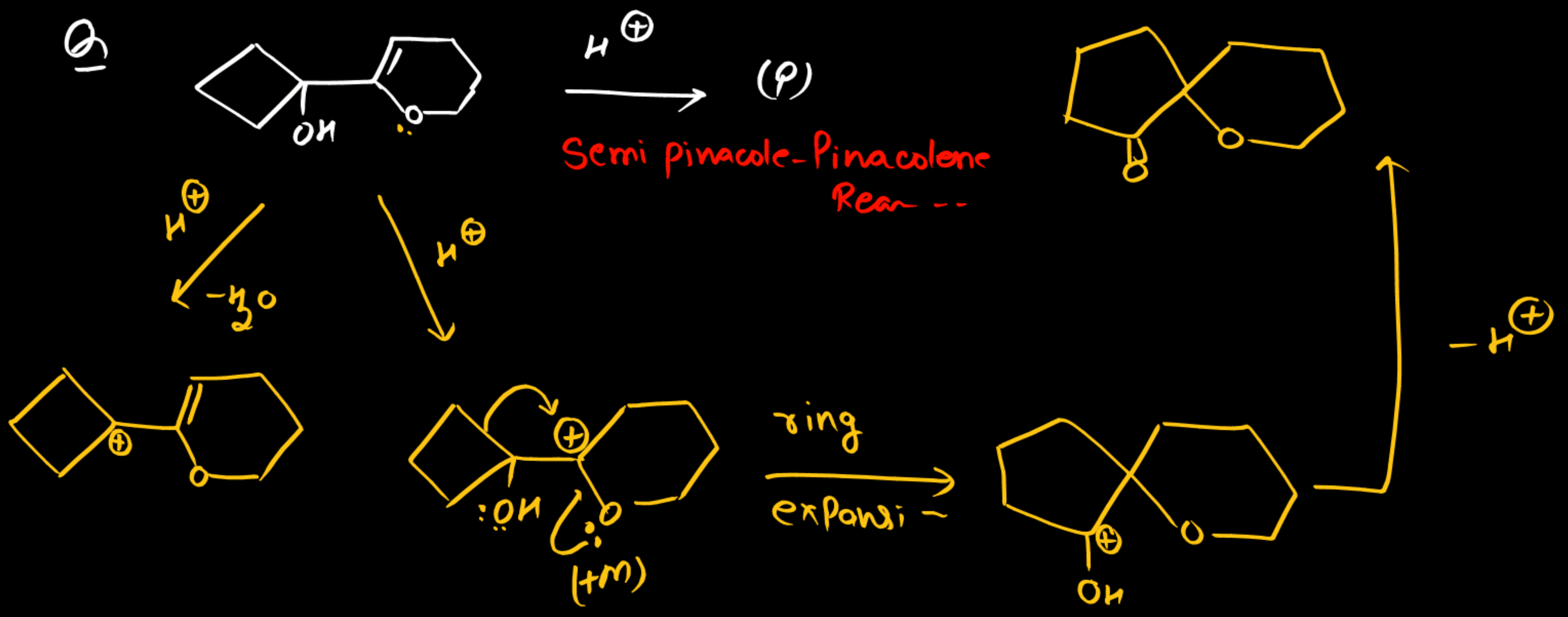
Q.


Br
HOBr
(X)
1 eq. NaHS
y
NaOH (1mole)

Sol

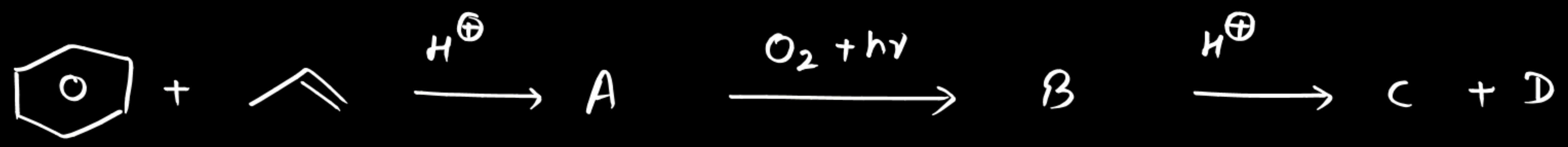

Br
HOBr
Br
Br
1°
Br
(Z)
Br
:OH
(X)
1eq NaHS
SN2
Br
HS
OH
(Y)
NaOH (1mole)
acid-Base Rxn
Br
OH
S
Strong Nu
intramolecular SN2
S
OH
(Z)
-SH > -OH acidic strength

Q



$H^{\oplus}$
(P)
Semi pinacole-Pinacolene Rear--
$H^{\oplus}$
$-H_2O$
$H^{\oplus}$
ring expansion
(+M)
$-H^{\oplus}$

Q.

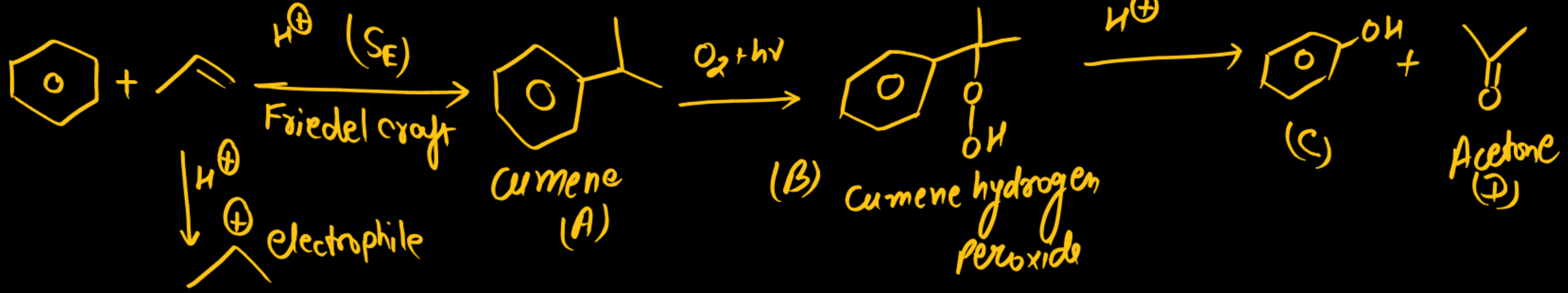


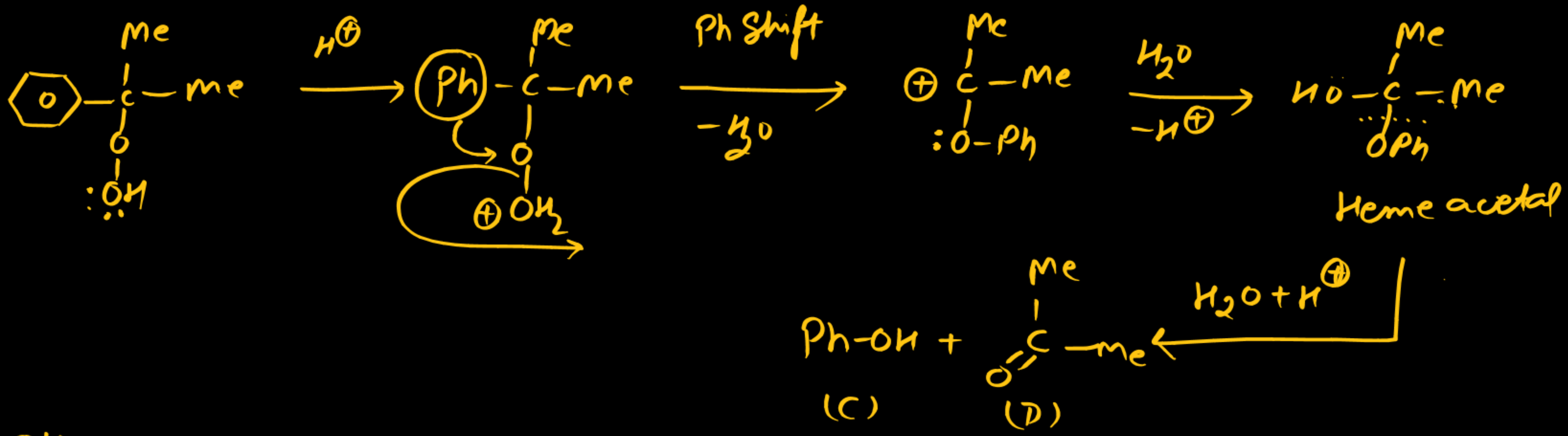

(C is more acidic than D)

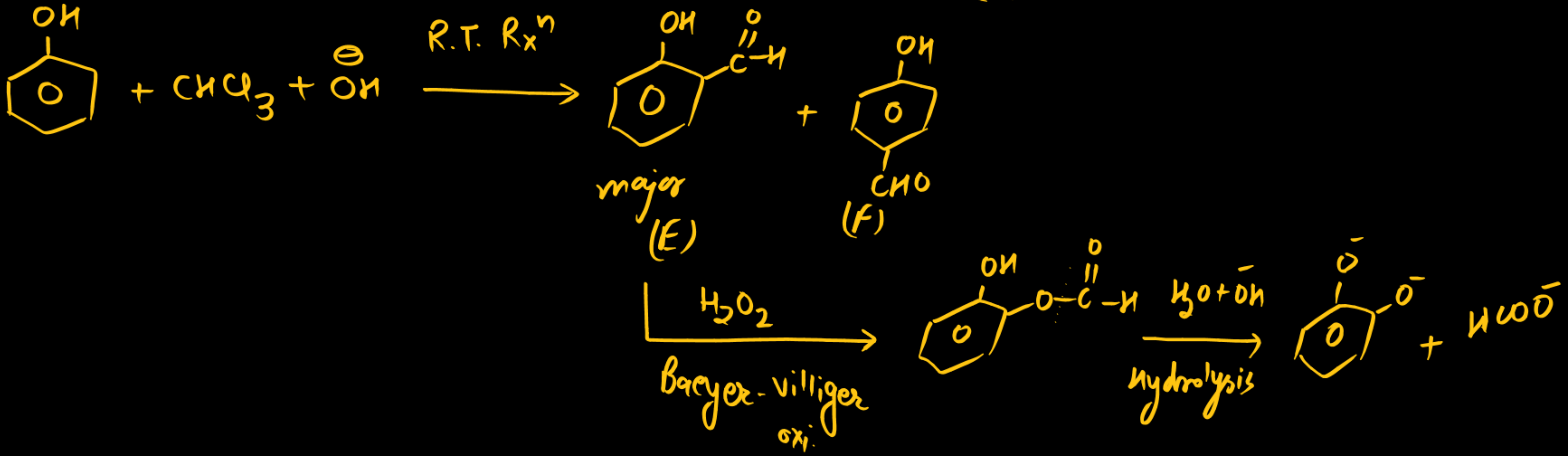
$$(C) \xrightarrow{CHCl_3 + KOH} \underset{\text{major}}{E} + F$$

$$E \xrightarrow{H_2O_2} G \xrightarrow{H_2O + \bar{O}H} H + I$$

Sol




$H^{\oplus}$
Ph shift
$-H_2O$
$H_2O$
$-H^{\oplus}$
Heme acetal
$H_2O + H^{\oplus}$
Ph-OH
(C)
(D)


R.T. Rx$^n$
major
(E)
(F)
$H_2O_2$
Baeyer-Villiger oxi.
$H_2O + \bar{O}H$
hydrolysis
$HCO\bar{O}$

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Q Compounds A, B and C are isomeric alcohols with formula $\underline{C_5H_{12}O}$ DU=0

A on oxidation gives ketone, B gives acid while C is not oxidised, A gives test with $I_2 + NaOH$.

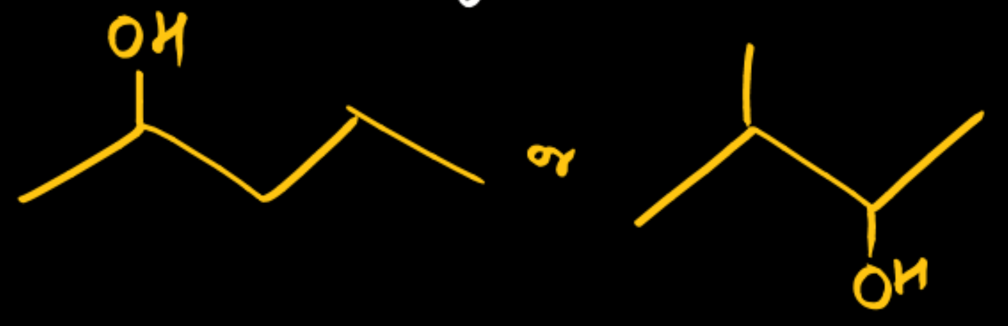
Give all possible structures of A, B, C.

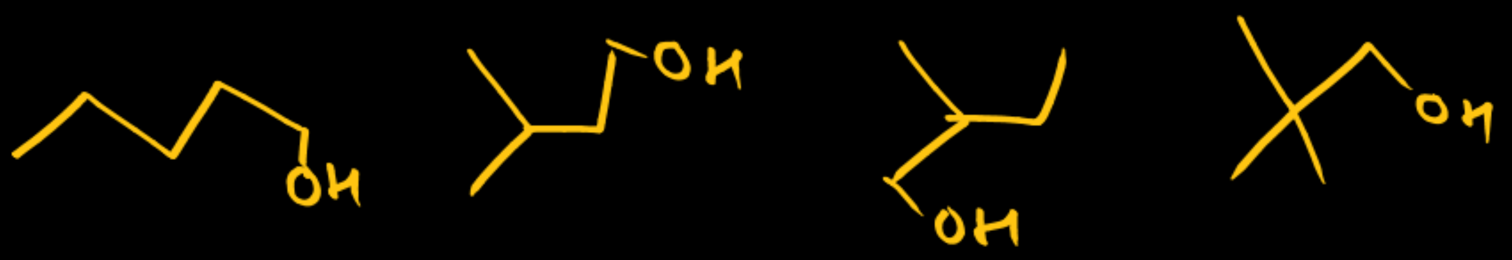
**Sol**

A 2° alcohol

B 1° alcohol

C 3° alcohol

A 

B 

C 

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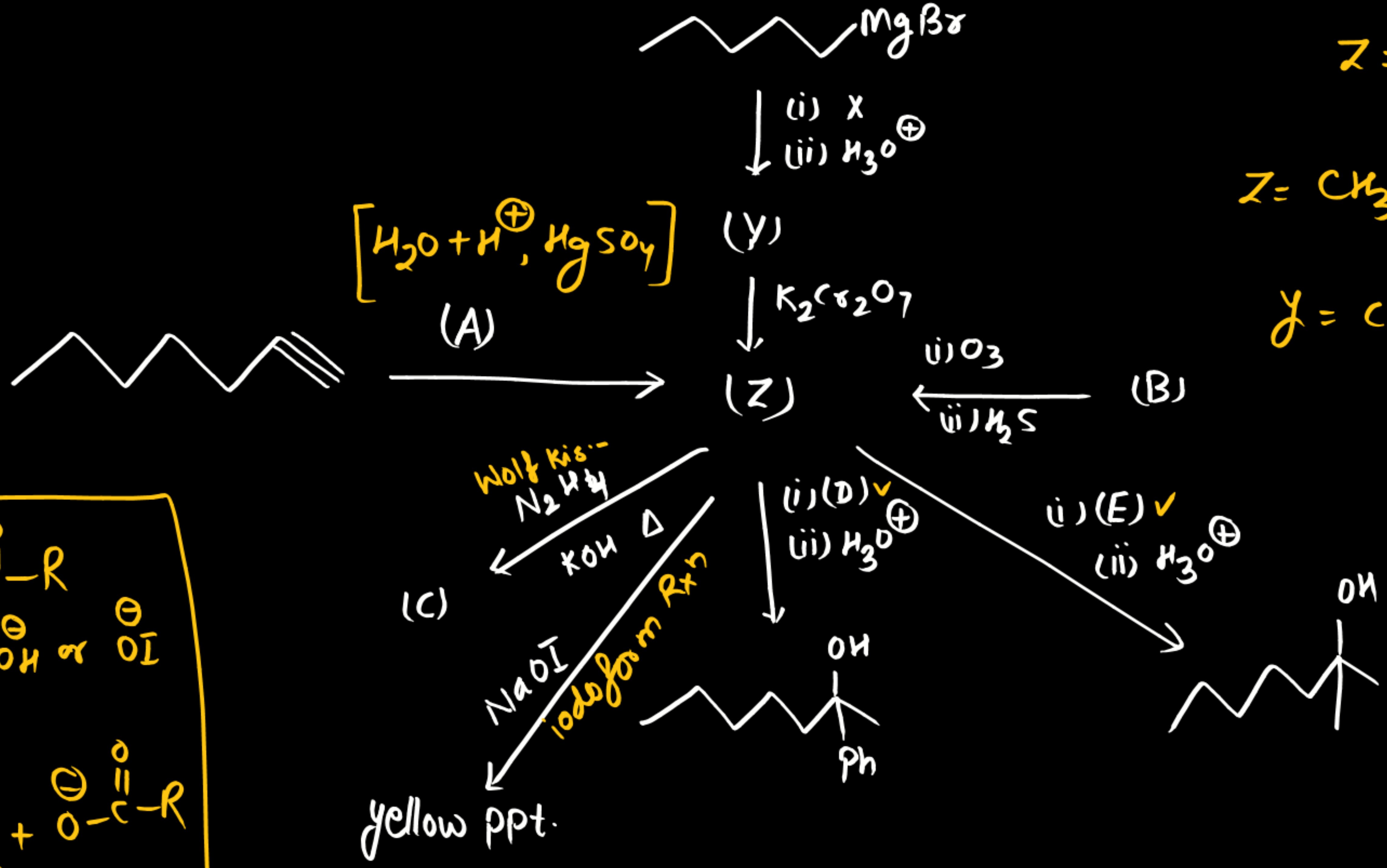
Q.




Z = Ketone / Aldehyde

$Z = CH_3-\overset{O}{\overset{\|}{C}}-R$

$Y = CH_3-\overset{OH}{\overset{|}{C}H}-R$ (2°)

$CH_3-\overset{O}{\overset{\|}{C}}-R$

$I_2 + \overset{\ominus}{O}H$ or $\overset{\ominus}{O}I$

$\downarrow$

$CHI_3 + \overset{\ominus}{O}-\overset{O}{\overset{\|}{C}}-R$

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